

Microsoft Sentinel SIEM/SOC Lab

Hybrid Endpoint Monitoring, Threat Detection, and Incident Response

Project Portfolio Report

Prepared By

Bikash Raya

Key Technologies

Microsoft Sentinel • Azure Arc • Azure Monitor Agent (AMA) • Log Analytics Workspace • Data Collection Rules (DCR) • Data Collection Endpoint (DCE) • Microsoft Azure • Microsoft Defender • VMware Workstation • Windows 11 • Ubuntu 22.04 LTS • Windows Server 2025

Project Scope

- Hybrid Endpoint Onboarding
 - Centralized Log Collection
 - Security Monitoring
 - Threat Detection
 - Incident Investigation and Response
-

June 2026

Microsoft Sentinel SIEM/SOC Lab.....	1
1. Executive Summary.....	3
2. Project Objectives.....	3
3. Solution Architecture.....	3
Core Components.....	3
Monitored Endpoints.....	4
Data Flow Architecture.....	4
4. Microsoft Sentinel Infrastructure Deployment.....	5
4.1 Resource Group Creation.....	5
4.2 Log Analytics Workspace Deployment.....	6
4.3 Data Collection Endpoint Configuration.....	9
4.4 Microsoft Sentinel Enablement.....	12
5. Role-Based Access Control (RBAC).....	13
6. Endpoint Deployment and Preparation.....	14
6.1 VMware Windows 11 Virtual Machine.....	14
6.2 VMware Ubuntu 22.04 LTS Virtual Machine.....	16
6.3 Azure Windows Server 2025 Virtual Machine.....	18
7. Azure Arc Onboarding.....	18
Onboarding Process:.....	19
8. Azure Monitor Agent and Data Collection Rules.....	23
8.1 Azure Monitor Agent Installation.....	23
8.2 Data Collection Rules (DCR) Configuration.....	24
Windows Data Sources.....	24
Linux Data Sources.....	24
9. Telemetry Validation.....	29
9.1 Heartbeat Validation.....	29
9.2 Syslog Validation.....	30
10. Threat Detection Configuration.....	31
10.1 Windows Security Auditing.....	32
10.2 Failed RDP Login Analytics Rule.....	32
11. Alert and Incident Validation.....	35
12. Incident Response.....	38
13. Results.....	39
14. Conclusion.....	40

1. Executive Summary

This project focused on designing and implementing a hybrid Security Information and Event Management (SIEM) laboratory using Microsoft Sentinel. The objective was to build a centralized monitoring environment capable of collecting, analyzing, and detecting security events across both on-premises and cloud-based systems.

The environment consisted of a Windows 11 virtual machine and an Ubuntu 22.04 virtual machine hosted in VMware Workstation to simulate on-premises infrastructure, along with an Azure-hosted Windows Server 2025 virtual machine. Azure Arc was used to onboard the on-premises systems into Azure, while Azure Monitor Agent (AMA), Data Collection Rules (DCR), and a Data Collection Endpoint (DCE) were configured to collect telemetry and forward logs to Microsoft Sentinel.

The project successfully demonstrated endpoint onboarding, centralized log management, telemetry validation, and threat detection through a custom analytics rule designed to identify failed Remote Desktop Protocol (RDP) login attempts.

2. Project Objectives

The key objectives of this project were:

The objectives of this project were to:

- Deploy Microsoft Sentinel as a centralized SIEM platform.
- Create and configure a Log Analytics Workspace for centralized log storage.
- Configure a Data Collection Endpoint (DCE) and Data Collection Rules (DCR).
- Implement Role-Based Access Control (RBAC) for secure administration.
- Onboard on-premises Windows and Linux virtual machines using Azure Arc.
- Integrate an Azure-hosted Windows Server 2025 virtual machine into the monitoring environment.
- Install and configure Azure Monitor Agent (AMA) on all endpoints.
- Validate telemetry ingestion through Heartbeat, Syslog, and Windows Security Event logs.
- Create a custom analytics rule for Failed RDP Login detection.
- Generate, investigate, and resolve security incidents within Microsoft Sentinel.

3. Solution Architecture

Core Components

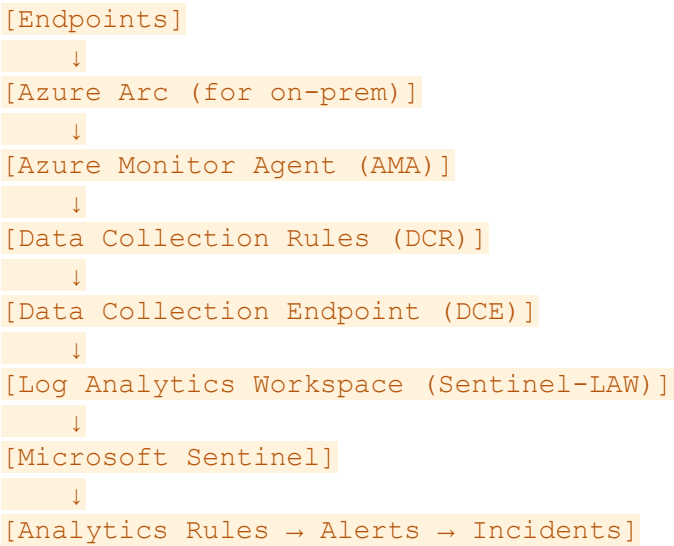
Component	Purpose
-----------	---------

Microsoft Sentinel	Central SIEM for threat detection, analytics, incident management, and investigation.
Log Analytics Workspace (Sentinel-LAW)	Centralized repository for all collected logs and telemetry.
Data Collection Endpoint (Sentinel-DCE)	Secure endpoint for agents to send data to Azure Monitor services.
Data Collection Rules (DCR)	Define what data is collected and where it is sent.
Azure Monitor Agent (AMA)	Lightweight agent installed on endpoints to collect and forward logs.
Azure Arc	Enables management of on-premises servers in Azure.

Monitored Endpoints

Endpoint	Type	Hosting Environment
Windows 11 VM	On-Premises	VMware Workstation
Ubuntu 22.04 LTS VM	On-Premises	VMware Workstation
Windows Server 2025 VM	Cloud	Microsoft Azure

Data Flow Architecture



A Data Collection Endpoint (Sentinel-DCE) was deployed to support Azure Monitor Agent communication and Data Collection Rule associations within the monitoring environment.

4. Microsoft Sentinel Infrastructure Deployment

4.1 Resource Group Creation

A dedicated resource group named **Sentinel-RG-Lab** was created to logically group and manage all related Azure resources.

Screenshots: Resource Group "Sentinel-RG-Lab" created in Azure Portal.

The first screenshot shows the 'Create a resource group' wizard in the Azure portal. The 'Basics' tab is selected, and the following information is entered:

- Subscription: Azure subscription 1
- Resource group name: Sentinel-RG-Lab
- Region: (Asia Pacific) Australia East

The 'Review + create' button is visible at the bottom of the wizard.

The second screenshot shows the 'Resource Manager | Resource groups' page in the Azure portal. The 'Sentinel-RG-Lab' resource group is listed under 'Azure subscription 1'. The 'Notifications' panel on the right shows a message: 'Resource group created' with the text 'Creating resource group 'Sentinel-RG-Lab' in subscription 'Azure subscription 1' succeeded.' and buttons to 'Go to resource group' and 'Pin to dashboard'.

4.2 Log Analytics Workspace Deployment

The **Sentinel-LAW** workspace was deployed within the **Sentinel-RG-Lab** resource group. This workspace serves as the foundation for log storage, query execution, and integration with Microsoft Sentinel.

Screenshots: Log Analytics Workspace "Sentinel-LAW" successfully deployed.

Resource Manager - Microsoft Azure

Home > Resource Manager

Resource Manager | Resource groups

Search resources, services, and docs (G+J)

How to manage changes with deployment tools? Export resource groups using Bicep or Terraform

Create Manage view Refresh Export to CSV Open query Assign tags Group by none

Filter for any field... Subscription equals all Location equals all Add filter

Name	Subscription
Sentinel-RG-Lab	Azure subscription 1

Showing 1 - 1 of 1. Display count: auto

Notifications

More events in the activity log → Dismiss all

Resource group created

Creating resource group 'Sentinel-RG-Lab' in subscription 'Azure subscription 1' succeeded.

[Go to resource group](#) [Pin to dashboard](#)

a few seconds ago

Create Log Analytics workspace

Home > Log Analytics workspaces

Create Log Analytics workspace

Basics Tags Review + Create

A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#)

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * Sentinel-RG-Lab [Create new](#)

Instance details

Name * Sentinel-LAW

Region * Australia East

[Review + Create](#) [Previous](#) [Next: Tags >](#)

Create Log Analytics workspace - X

+

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Help + support

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

BikSecOps_Labs@outlo...
DEFAULT DIRECTORY (BMSACOPS)

Home > Log Analytics workspaces

Create Log Analytics workspace ...

✓ Validation passed

Basics Tags Review + Create

Log Analytics workspace

by Microsoft

Basics

Subscription Azure subscription 1

Resource group Sentinel-RG-Lab

Name Sentinel-LAW

Region Australia East

Pricing

Pricing tier Pay-as-you-go (Per GB 2018)

The cost of your workspace depends on the volume of data ingested and how long it is retained. Regional pricing details are available on the [Azure Monitor pricing page](#). You can change to a different pricing tier after the workspace is created. [Learn more about Log Analytics pricing models.](#)

Tags

None

Create

Previous

Download a template for automation

17°C
Light rain

Search

2:34 P
2/06/2025

The top screenshot displays the 'Microsoft.LogAnalyticsOMS | Overview' page in the Azure portal. A red box highlights the 'Your deployment is complete' message, which includes the deployment name 'Microsoft.LogAnalyticsOMS', subscription 'Azure subscription 1', and resource group 'Sentinel-RG-Lab'. Below this, the 'Deployment details' table shows the 'Sentinel-LAW' resource as a 'Log Analytics workspace' with a status of 'OK'. The 'Next steps' section includes a 'Go to resource' button.

The bottom screenshot shows the 'Sentinel-LAW' Log Analytics workspace overview page. It includes a search bar, a list of resources, and a detailed view of the workspace. The 'Essentials' section shows the resource group 'sentinel-rg-lab', workspace name 'Sentinel-LAW', and other details. The 'Get started with Log Analytics' section provides instructions on how to connect data sources, configure monitoring solutions, and monitor workspace health.

4.3 Data Collection Endpoint Configuration

A **Data Collection Endpoint (Sentinel-DCE)** was created to act as the destination for telemetry from Azure Monitor Agents. The endpoint uses HTTPS and supports secure communication from hybrid environments.

Additionally, the **Microsoft.Insights** resource provider was registered to enable monitoring features.

In Azure Cloud Shell:

```
az provider register --namespace Microsoft.Insights
```

Check status:

```
az provider show --namespace Microsoft.Insights --query registrationState
```

Wait until output is:

"Registered"

Screenshot: DCE "Sentinel-DCE" created.

Microsoft Azure

Home > Data collection endpoints

Create data collection endpoint

Basics Tags Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all of your resources. [Learn more](#)

Endpoint details

Endpoint Name * Sentinel-OCE

Subscription * Azure subscription 1

Resource Group * Sentinel-RG-Lab

Region * Australia East

Review + create

Basics

Endpoint Name Sentinel-OCE

Subscription Azure subscription 1

Resource Group Sentinel-RG-Lab

Azure CLI

Run the following command:

```
Bash
az provider register --namespace Microsoft.Insights
```

To check the registration status, use:

```
Bash
az provider show --namespace Microsoft.Insights --query "registrationState"
```

Azure PowerShell

How do I check if a resource provider is registered in Azure?

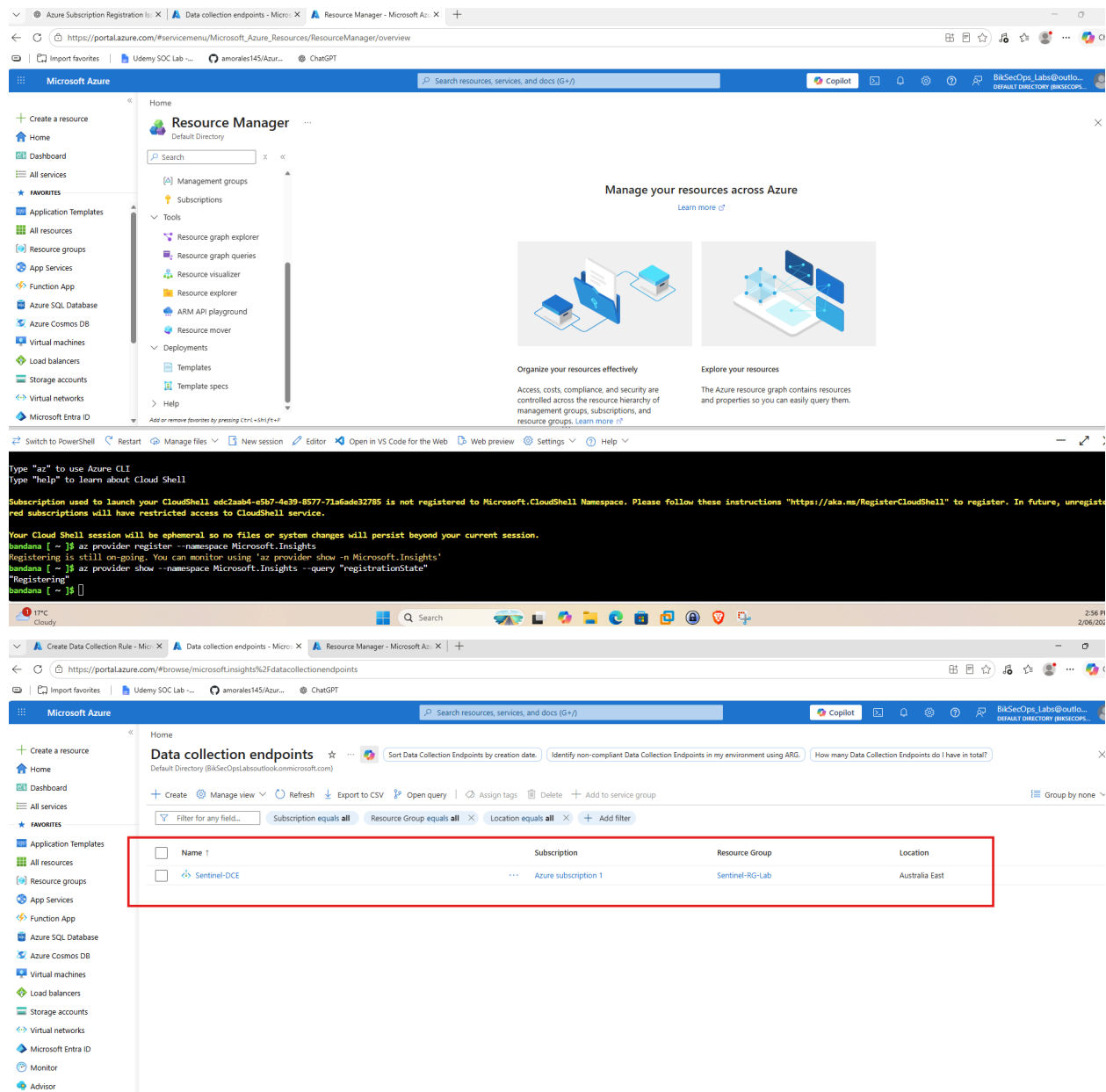
What permissions are needed to register a resource provider in Azure?

Can you guide me through registering a resource provider using Azure CLI?

I want to ...

Create < Previous Next >

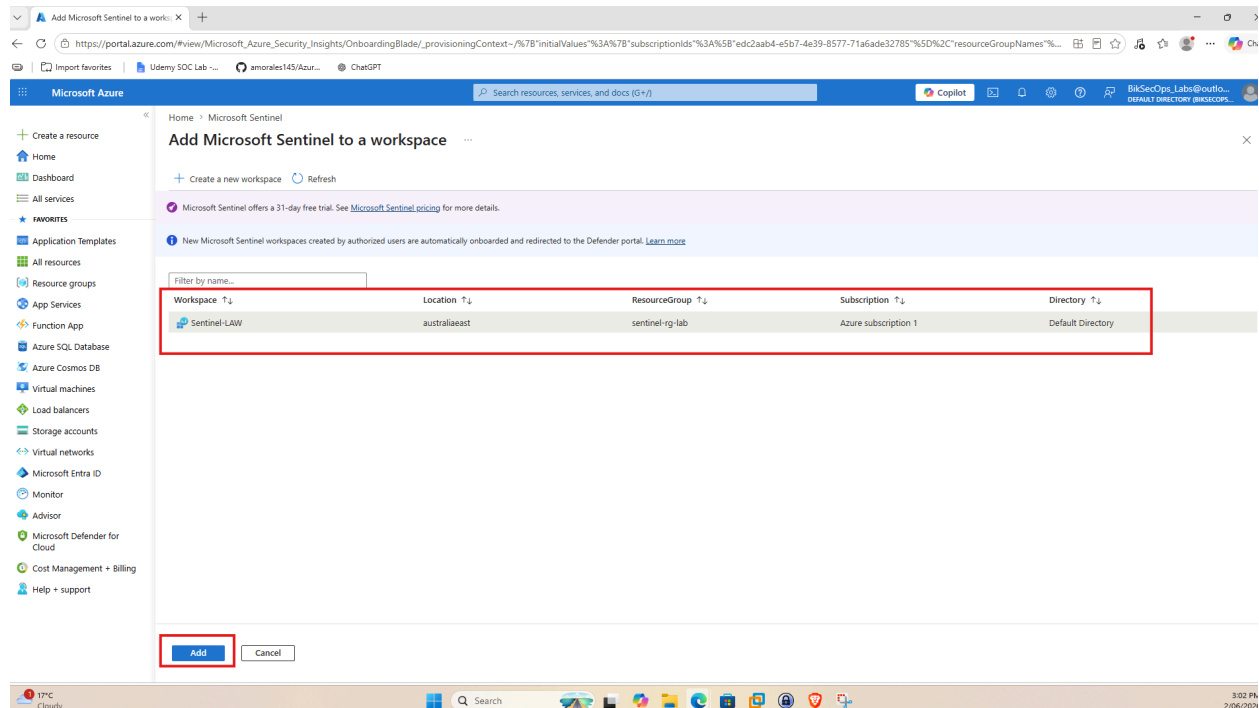
Screenshot: Microsoft.Insights resource provider registered.



4.4 Microsoft Sentinel Enablement

Microsoft Sentinel was enabled on the **Sentinel-LAW** workspace. This activated SIEM capabilities such as analytics rules, threat intelligence, incident management, and integration with other Microsoft security tools.

Screenshot: Microsoft Sentinel successfully enabled on the workspace.



5. Role-Based Access Control (RBAC)

To enforce least-privilege access, the following Azure roles were assigned to the administrator account:

Role	Scope	Purpose
Monitoring Reader	Resource Group	Allows viewing of monitoring data (e.g., logs, metrics).
Log Analytics Contributor	Workspace	Grants read/write access to Log Analytics data and configurations.
Microsoft Sentinel Contributor	Workspace	Enables creation and management of analytics rules, incidents, and playbooks.

These roles ensure that users can monitor and manage security operations without excessive permissions.

Screenshots: Role assignments applied to user.

The first screenshot shows the Microsoft Azure portal interface for the 'Sentinel-RG-Lab' resource group. The 'Access control (IAM)' page is selected, and the 'Add role assignment' button is highlighted with a red box. Below this, there are sections for 'Grant access to this resource', 'View access to this resource', 'View deny assignments', and 'Create a custom role'.

The second screenshot shows the 'Add role assignment' page. The 'Role' tab is selected, and the 'Monitoring Reader' role is highlighted with a red box. The 'Next' button at the bottom is also highlighted with a red box.

Name	Description	Type	Category	Details
Azure Monitor Pipeline Reader	Read Azure Monitor Pipeline resources, including pipeline configurations and monitoring features.	BuiltInRole	None	View
Log Analytics Reader	Log Analytics Reader can view and search all monitoring data as well as view monitoring settings, including viewing the configuratio...	BuiltInRole	Analytics	View
Monitoring Data Reader	Can access the data in an Azure Monitor Workspace.	BuiltInRole	None	View
Monitoring Reader	Can read all monitoring data.	BuiltInRole	Monitor	View
Privileged monitoring data reader	Privileged monitoring data reader can query and search the logs it is allowed to view over Log Analytics workspaces and tables, includin...	BuiltInRole	None	View

6. Endpoint Deployment and Preparation

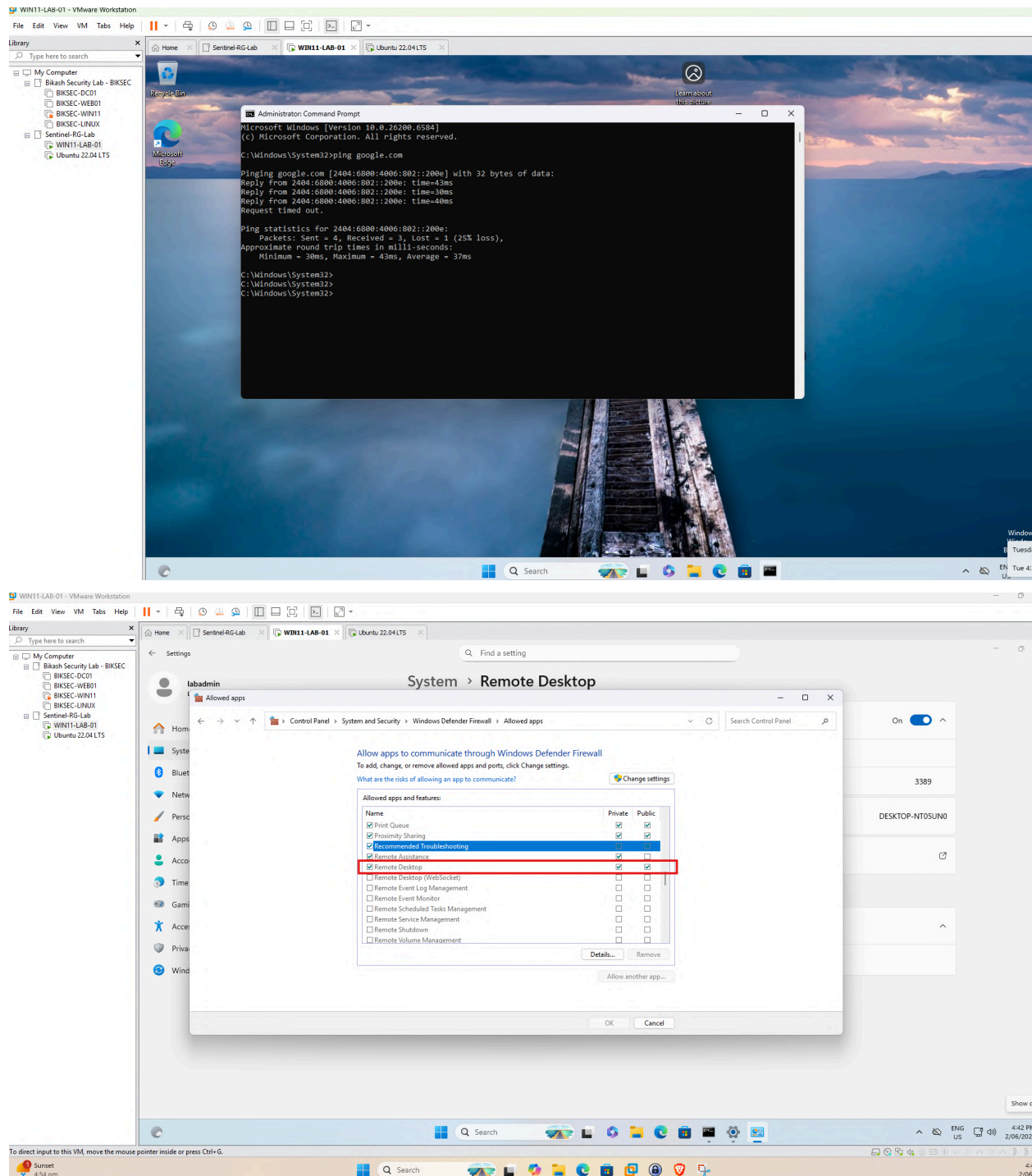
6.1 VMware Windows 11 Virtual Machine

A **Windows 11 Pro** VM was deployed on VMware Workstation to simulate an on-premises workstation.

Configuration Steps:

- Enabled **Remote Desktop Protocol (RDP)**.
- Configured **Windows Firewall** to allow RDP and outbound HTTPS.
- Verified network connectivity to Azure.
- Tested local event generation.

Screenshot: Windows 11 VM running in VMware.



6.2 VMware Ubuntu 22.04 LTS Virtual Machine

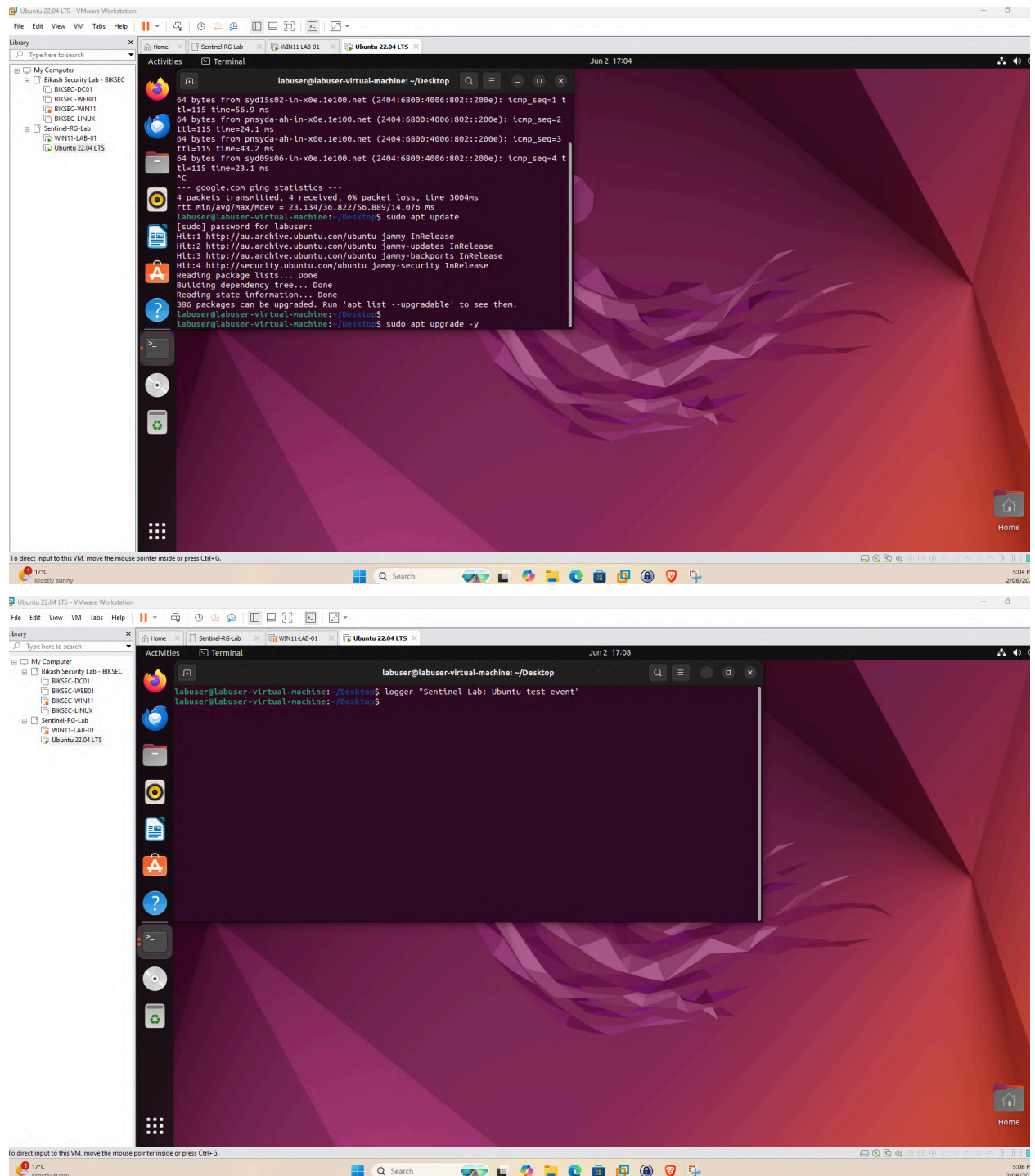
An Ubuntu 22.04 LTS Server VM was deployed to represent a Linux server in a hybrid environment.

Configuration Steps:

- Applied system updates: `sudo apt update && sudo apt upgrade -y`

- Installed `rsyslog` (enabled by default).
- Generated test syslog entries to validate logging functionality.
- Ensured outbound connectivity to Azure endpoints.

Screenshot: Ubuntu VM.

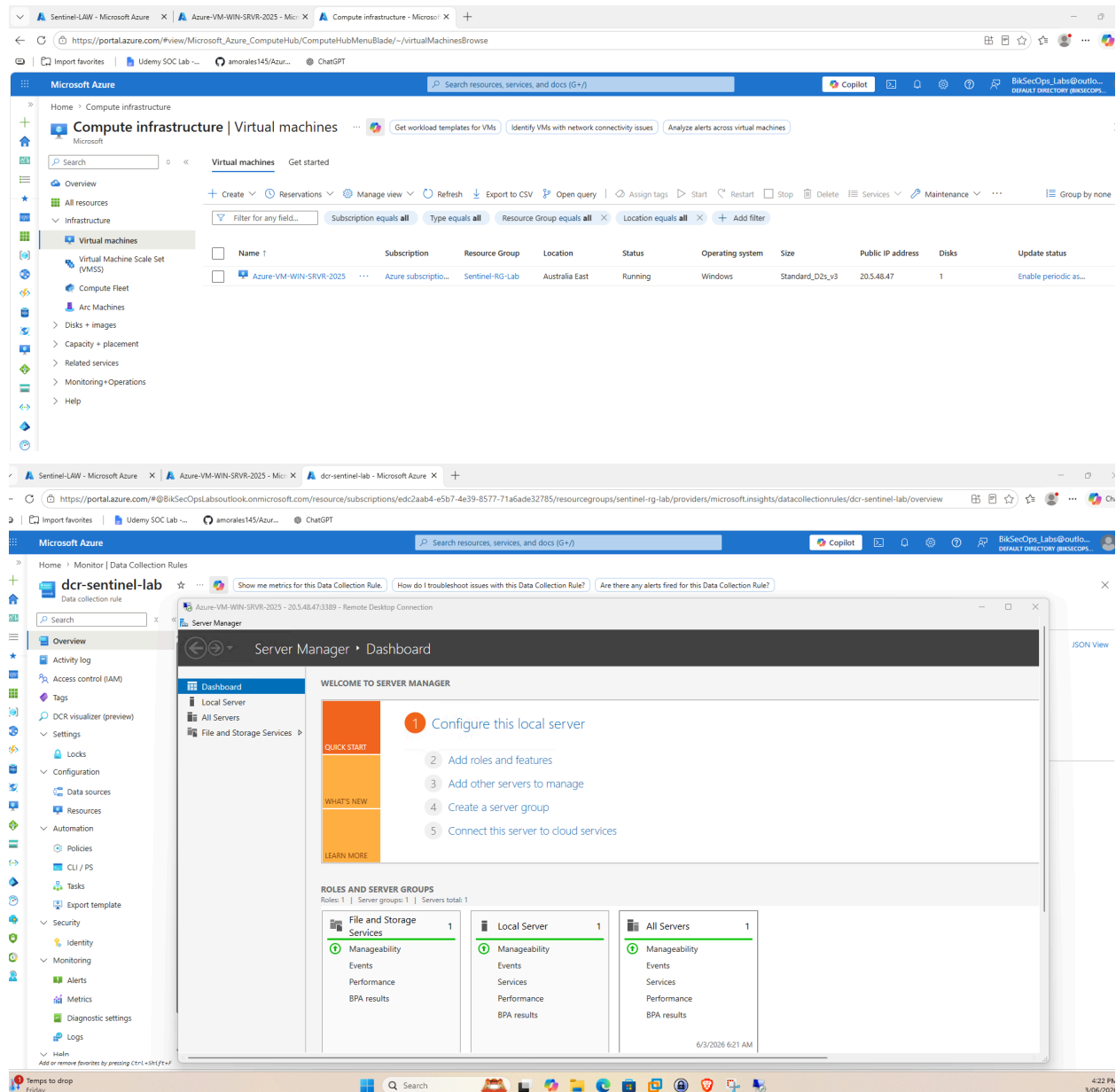


6.3 Azure Windows Server 2025 Virtual Machine

A Windows Server 2025 Datacenter VM was provisioned in Azure to simulate a cloud workload.

- Deployed with **System-Assigned Managed Identity**.
- Public IP assigned for RDP access during setup.
- Joined to the same Log Analytics Workspace for telemetry ingestion.

Screenshot: Azure Portal view of Windows Server 2025 VM.



7. Azure Arc Onboarding

Azure Arc enabled the extension of Azure management to non-Azure machines.

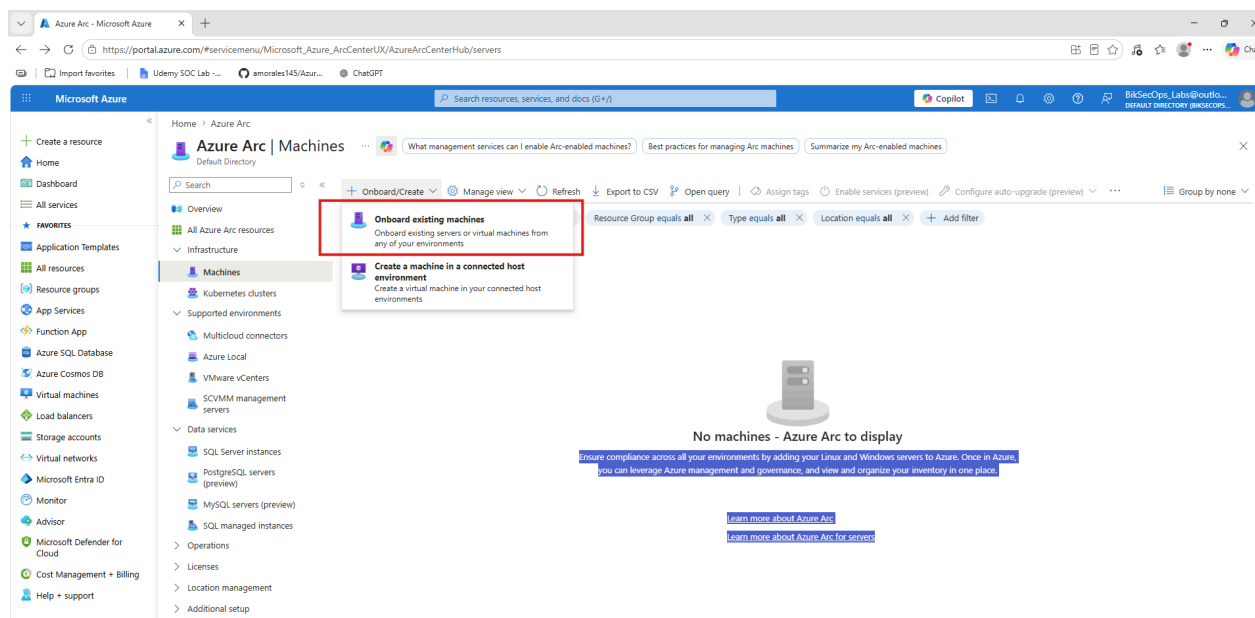
Onboarding Process:

1. Generated **onboarding script** from the Azure Portal under **Azure Arc > machines > create> onboard** and complete the steps to download the script
2. Executed the script on both the **Windows 11** and **Ubuntu** VMs.
 1. Run the commands to fix the issue running the script file :PS
C:\WINDOWS\system32> Get-ExecutionPolicy RemoteSigned
PS C:\WINDOWS\system32> Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
3. Script automatically:
 1. Installed the **Azure Connected Machine Agent**.
 2. Authenticated with Azure using service principal or interactive login.
 3. Registered the machine as an Azure resource.

After successful onboarding:

- Both VMs appeared in the Azure Portal under **Azure Arc > Servers**.
- Tags and resource metadata were applied for organization.

Screenshots: Azure Arc > machines > create> Onboard existing machines.



The image displays two screenshots of the Microsoft Azure portal interface, specifically the 'Onboard existing machines with Azure Arc' wizard.

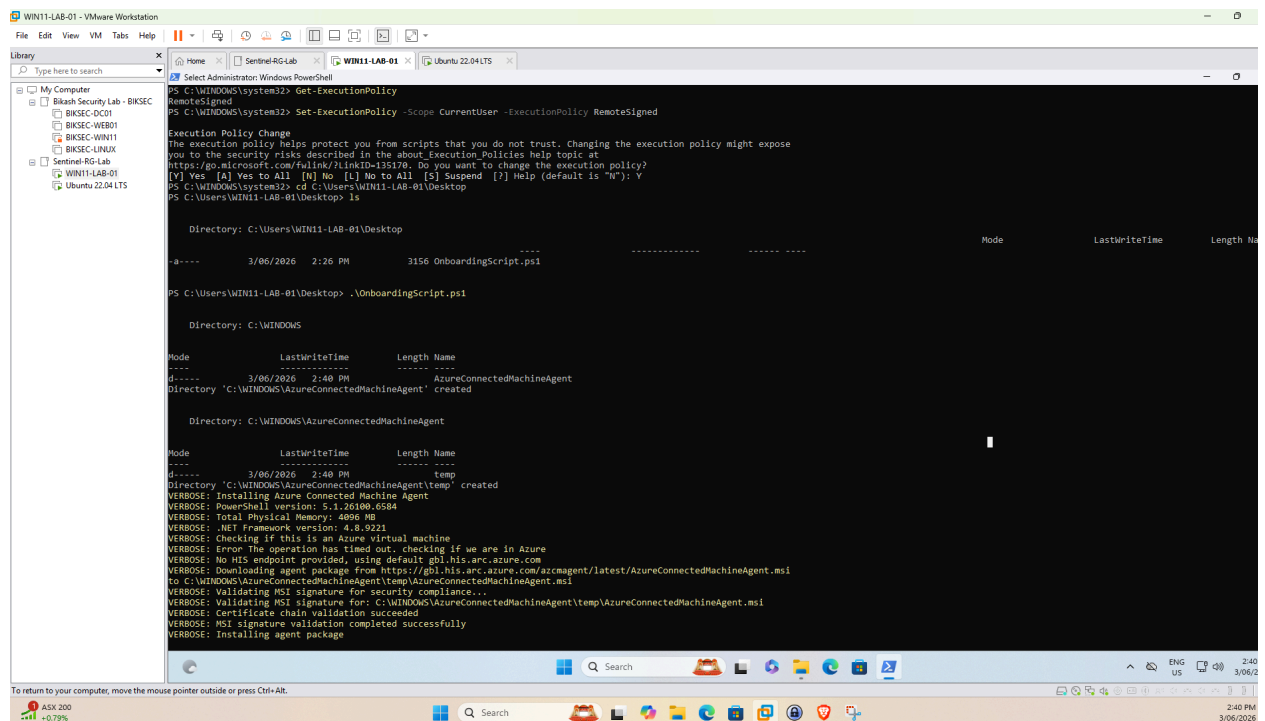
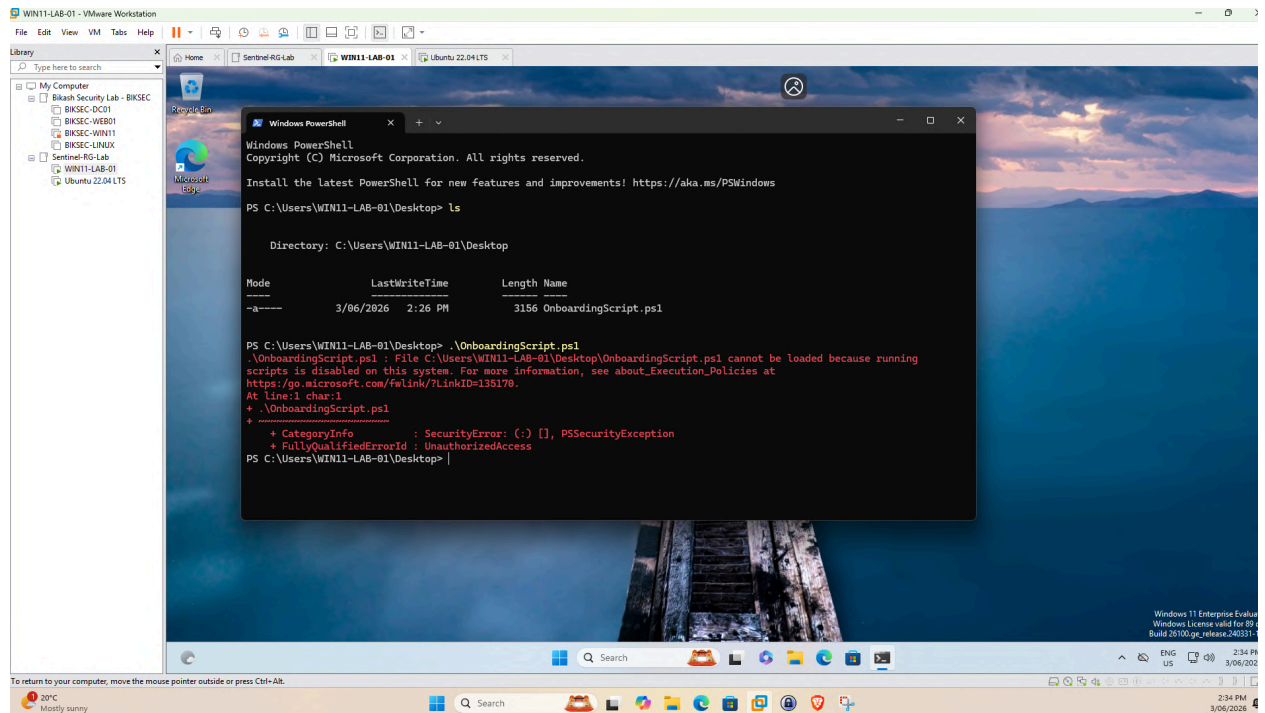
Top Screenshot (Basics tab):

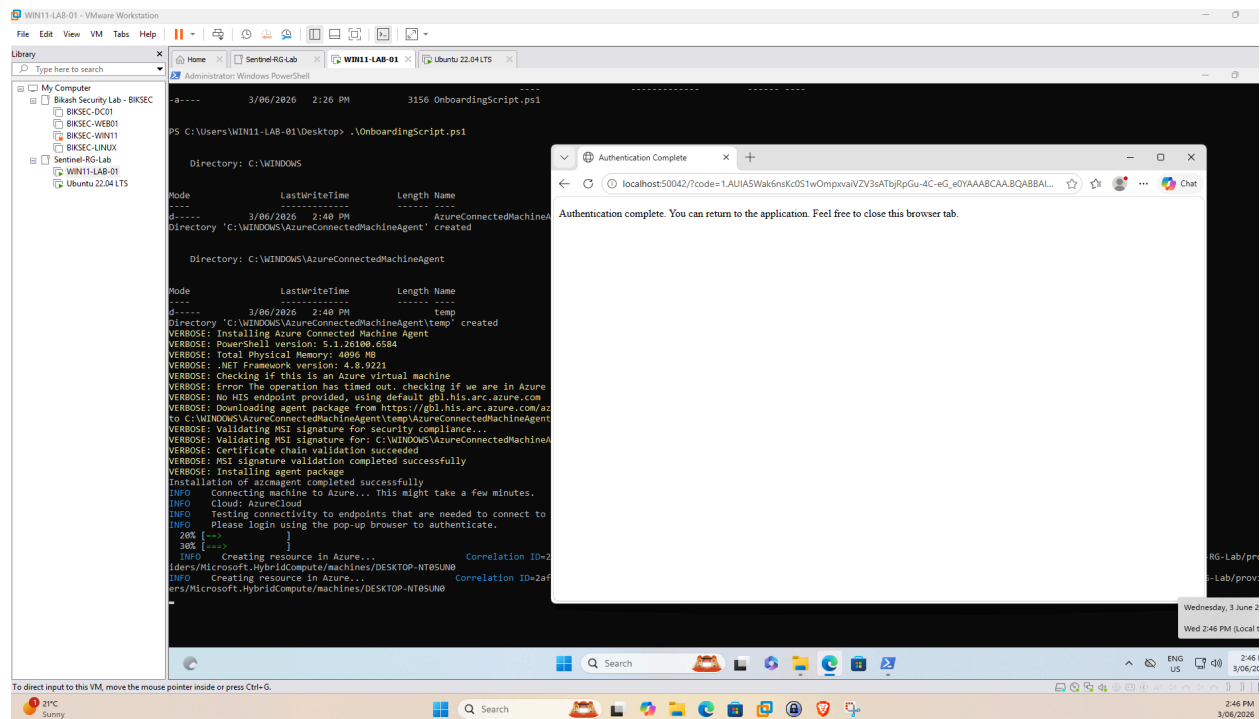
- Page Title:** Onboard existing machines with Azure Arc
- Navigation:** Basics, Tags, Download and run script
- Project details:** Select the subscription and resource group where you want the server to be managed within Azure.
 - Subscription: Azure subscription 1
 - Resource group: Sentinel-RG-Lab
- Server details:** Select details for the servers that you want to add. An agent package will be generated for the selected server type.
 - Region: (Asia Pacific) Australia East
 - Operating system: Windows
- SQL Server:** Connect SQL Server ☒
 - Automatically connect any SQL Server instances to Azure Arc. [Learn more](#)
- Buttons:** < Previous, Next >, Download and run script

Bottom Screenshot (Download and run script tab):

- Page Title:** Onboard existing machines with Azure Arc
- Navigation:** Basics, Tags, Download and run script
- Section:** 1. Download or copy the following script
- Script Content:** A PowerShell script for downloading and running the Azure Arc agent. The script includes comments and logic for handling permissions and environment variables.
- Buttons:** Download, Close

Screenshots: Issue and fix for running script file in Win 11 VM





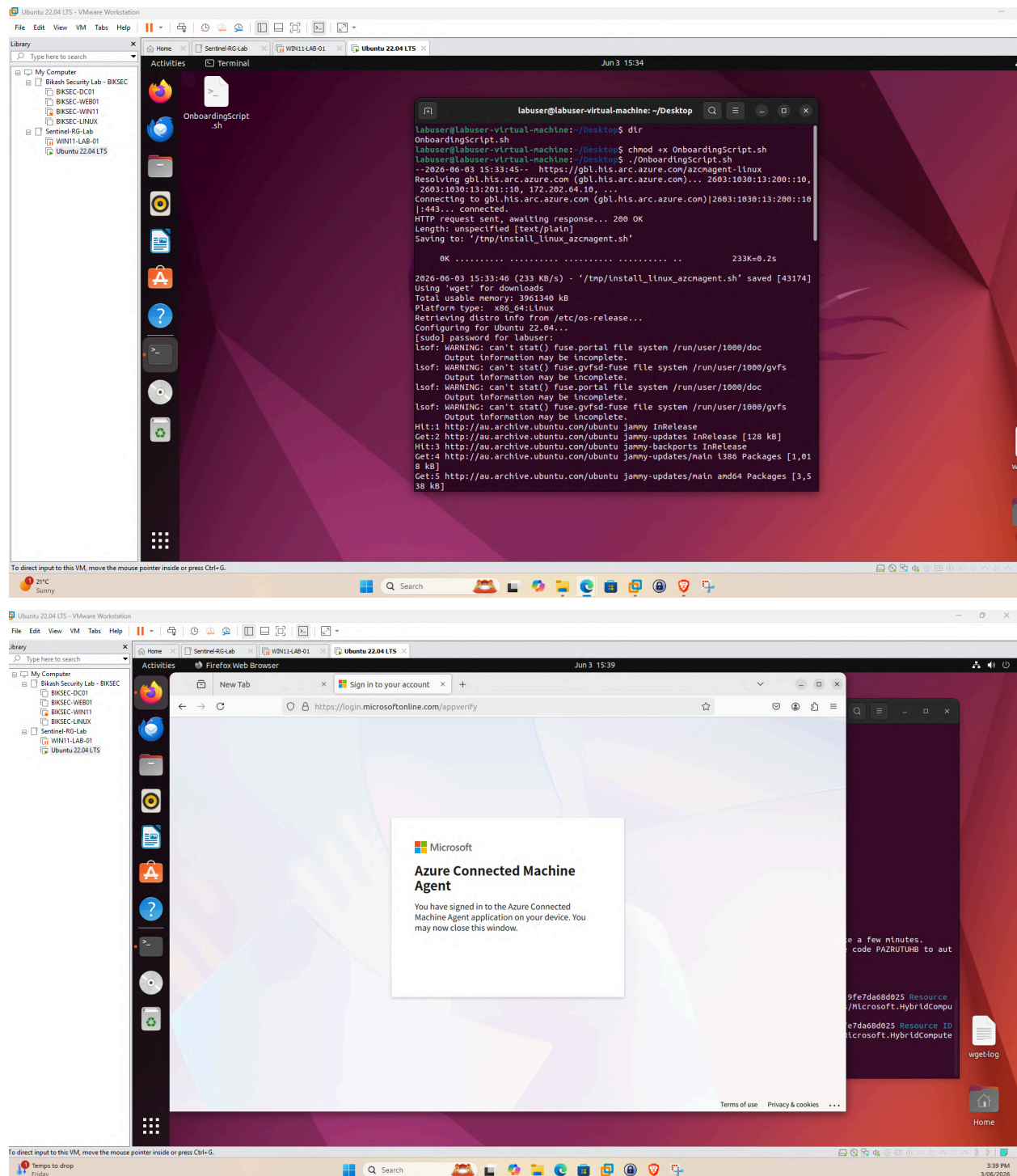
Screenshot: Ubuntu VM successfully connected via Azure Arc.

labuser@labuser-virtual-machine :~ /Desktop\$ dir

OnboardingScript.sh

labuser@labuser-virtual-machine :~ /Desktop\$ chmod +x OnboardingScript. sh

labuser@labuser-virtual-machine :~ /Desktop\$./OnboardingScript. sh



8. Azure Monitor Agent and Data Collection Rules

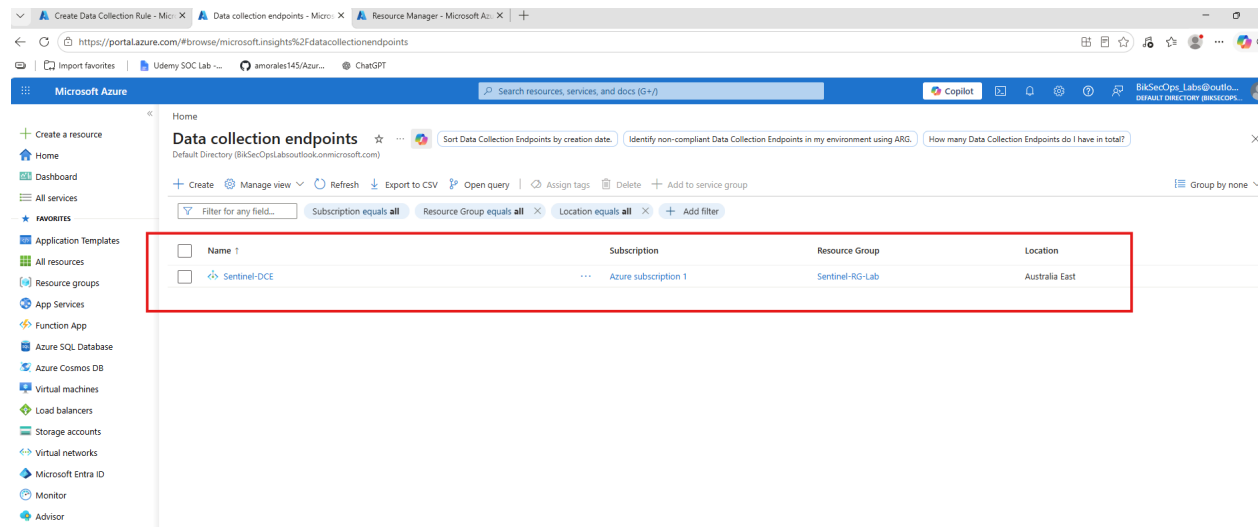
8.1 Azure Monitor Agent Installation

The **Azure Monitor Agent (AMA)** was installed on all three endpoints:

Endpoint	Installation Method
Windows 11 VM	PowerShell script
Ubuntu VM	Bash script via terminal
Azure Windows Server 2025	Azure Extension (automated)

The agent establishes secure TLS connections to the **Data Collection Endpoint (Sentinel-DCE)** and begins transmitting logs based on DCR associations.

Screenshot: Sentinel-DCE



8.2 Data Collection Rules (DCR) Configuration

Two DCRs were created:

1. **Windows-DCR:** Collects Windows Event Logs.
2. **Linux-DCR:** Collects Syslog messages.

Windows Data Sources

- **Event Types:** Security
- **Event IDs Included:** 4624 (Success), 4625 (Failure), 4648 (Explicit Credentials), etc.
- **Channels:** Security, System, Application

Linux Data Sources

- **Facilities:** auth, authpriv, cron, daemon, kern, user
- **Log Levels:** info, notice, warning, err, crit, alert, emerg

Each DCR was associated with its respective endpoints using **Data Collection Rule Associations (DCRA)**.

Screenshots: DCR configuration showing Windows and Linux Syslog sources.

Microsoft Azure

Home

Dashboard

All services

FAVORITES

Application Templates

All resources

Resource groups

App Services

Function App

Azure SQL Database

Azure Cosmos DB

Virtual machines

Load balancers

Storage accounts

Virtual networks

Microsoft Entra ID

Monitor

Advisor

Microsoft Defender for Cloud

Cost Management + Billing

Help + support

Home > Microsoft Sentinel | Data connectors > Content hub > Windows Security Events

Windows Security Events via AMA

Disconnected

Microsoft

Last Log Received

alerts, and improve investigation. This gives you more insight into your organization's network and improves your security operation capabilities.

Last data received

Related content

0 Workbooks

1 Queries

21 Analytics rules templates

Go to log analytics

Data received

SecurityEvents

0

Data types

SecurityEvents

Prerequisites

To integrate with Windows Security Events via

Workspace data sources: read and write per

To collect data from non-Azure VMs, they mu

Configuration

Enable data collection rule

Security Events logs are collected only from W

Refresh

Rule name

Created t

No results

+ Create data collection rule

Create Data Collection Rule

Data collection rule management

Basic

Resources

Collect

Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all of your resources.

Rule details

Rule name

Subscription

Resource group

DCR-WindowsSecurityEvents

Azure subscription 1

Sentinel-RG-Lab

Need Resources >

18°C

Partly sunny

Search

2:01 PM

4/06/2028

Microsoft Azure portal interface showing the "Add new data source" step for a Data Collection Rule. The "Data source" tab is selected, and the "Data source type" is set to "Windows Event Logs". The "Event Log Selection Mode" is set to "Basic (Use this to enable collection of event logs)". A list of event log categories is shown, including Application, Security, and System, with checkboxes for Critical, Error, Warning, Information, and Verbose. The "Save" button is visible at the bottom.

Microsoft Azure portal interface showing the "Add new data source" step for a Data Collection Rule. The "Destination" tab is selected, and the "Data collection rule region" is set to "Australia East". The "Add destination" button is highlighted. The "Destination type" is set to "Log Analytics Workspaces". The "Subscription" is set to "Azure subscription 1". The "Log Analytics Workspaces" dropdown is set to "Sentinel-LAW (Sentinel-RG-Lab)". The "Apply" button is visible at the bottom.

Windows Security Events via AMA

Disconnected Status Microsoft Provider Last Log Received

alerts, and improve investigation. This gives you more insight into your organization's network and improves your security operation capabilities.

Last data received

Related content

0 Workbooks 1 Queries 21 Analytics rules templates

Data received

Go to log analytics

SecurityEvents

Data types

SecurityEvents

Create Data Collection Rule

Data collection rule management

Basic Resources Collect Review + create

Select which events to stream.

All Security Events Common Minimal Custom

To integrate with Windows Security Events via AMA, you need to:

- Workspace data sources: read and write permissions on the data source.
- To collect data from non-Azure VMs, they must be connected to the workspace.

Configuration

Enable data collection rule

Security Events logs are collected only from Windows Security Events.

Refresh

Rule name Created time

No results

+ Create data collection rule

< Previous Next: Review + create >

Windows Security Events via AMA

Disconnected Status Microsoft Provider Last Log Received

alerts, and improve investigation. This gives you more insight into your organization's network and improves your security operation capabilities.

Last data received

Related content

0 Workbooks 1 Queries 21 Analytics rules templates

Data received

Go to log analytics

SecurityEvents

Data types

SecurityEvents

Create Data Collection Rule

Data collection rule management

Validation passed

Basic Resources Collect Review + create

Basic

Data rule name DCR-WindowsSecurityEvents

Subscription Azure subscription 1

Resource Group Sentinel-RG-Lab

Selected resources

Name	Type
desktop-nt05un0	microsoft.hybridcompute/machines

Selected events AllEvents

< Previous Create

▼

Add new data source - Microsoft | X

Monitor - Microsoft Azure | X

+

←

↻

https://portal.azure.com/#view/Microsoft_Azure_Monitoring/ConfigureDataflow.ReactView/IsNewDcr~/true/IsOpenedByKnockoutBlade~/false/editorMode/Add/subsriptionId/edc2aab4-e5b7-4e39-8577-71afade32785/locationId/australiaeast...

Import favorites

Udemy SOC Lab ~...

amores145/Azur...

ChatGPT

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

🔔

🔗

🔍

🗺️

👤

BikSecOps_Labs@outlo...
DEFAULT DIRECTORY (BIKSECOPS)

Home > Monitor > Data Collection Rules > Create Data Collection Rule

Add new data source ...

Data source

Destination

☐

Facility

☐

LOG_ALERT

☐

LOG_AUDIT

☒

LOG_AUTH

☒

LOG_AUTHPRIV

☐

LOG_CLOCK

☐

LOG_CRON

☒

LOG_DAEMON

☐

LOG_FTP

☒

LOG_KERN

☐

LOG_LOCAL0

☐

LOG_LOCAL1

☐

LOG_LOCAL2

☐

LOG_LOCAL3

Log Level

LOG_DEBUG

LOG_INFO

LOG_INFO

LOG_INFO

LOG_DEBUG

LOG_DEBUG

LOG_NOTICE

LOG_DEBUG

LOG_INFO

LOG_DEBUG

LOG_DEBUG

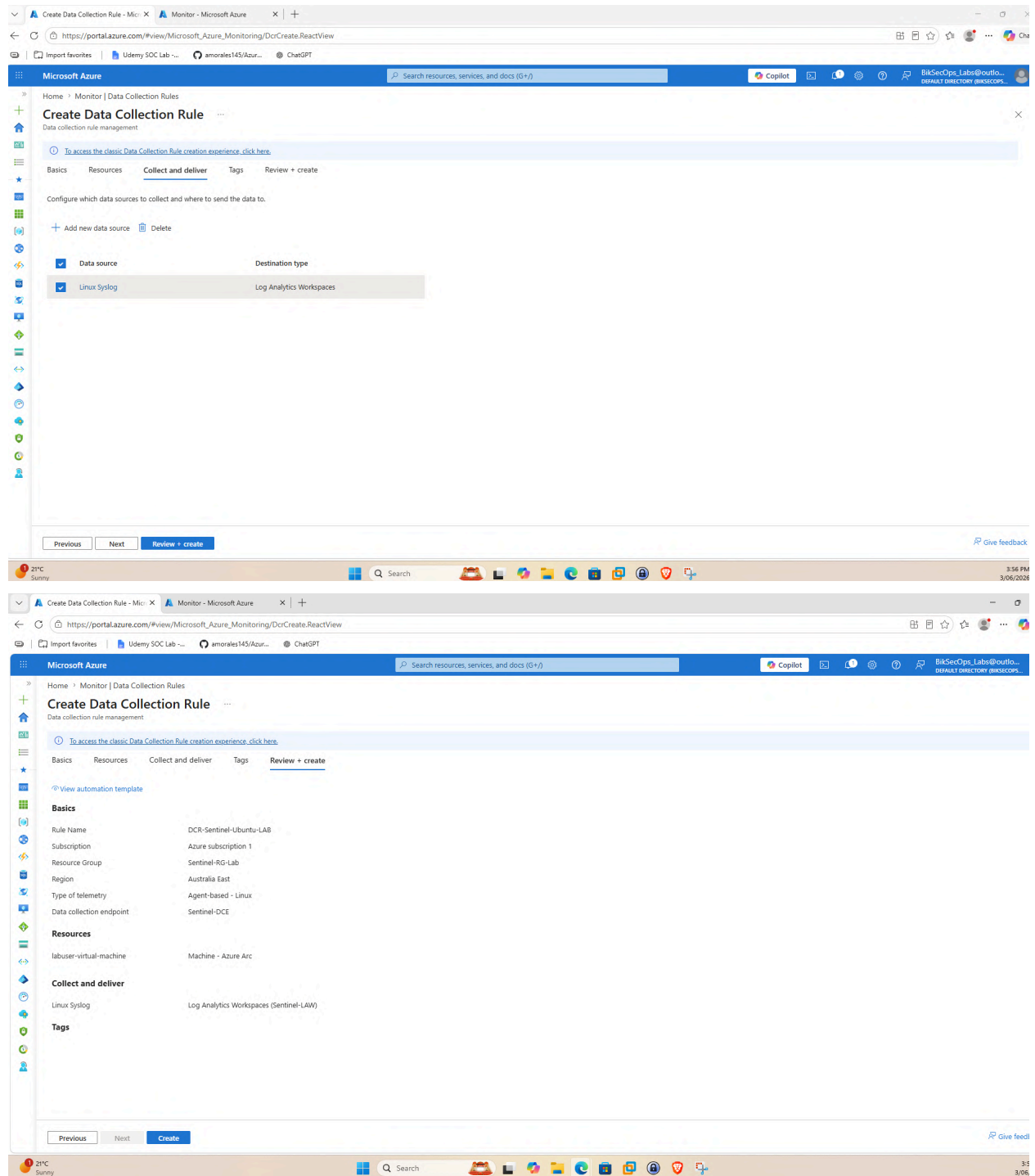
LOG_DEBUG

Save

Next: Destinations >

Cancel

Feedback



9. Telemetry Validation

9.1 Heartbeat Validation

Heartbeat logs confirm that agents are communicating with Azure Monitor.

KQL Query:

Heartbeat

```
| project TimeGenerated, Computer, OSType, OSName
| summarize LastSeen=max(TimeGenerated), OS=any(OSType) by Computer
| sort by LastSeen desc
```

Results:

- All three endpoints reported heartbeat data within the last 5 minutes.
- OS types correctly identified (Windows/Linux).
- Confirmed bidirectional communication.

Screenshot: Query results showing all three machines reporting Heartbeat.

The screenshot shows the Microsoft Azure Sentinel-LAW Logs interface. A KQL query is entered in the 'New Query 1*' field:

```
1 heartbeat |
2 project TimeGenerated, Computer, OSType, OSName |
3 summarize LastSeen = max(TimeGenerated), OS = any(OSType)
4 by Computer
5 | sort by LastSeen desc
```

The query results are displayed in a table with columns: Computer, LastSeen [UTC], and OS. The results show three machines reporting heartbeat data within the last 5 minutes.

Computer	LastSeen [UTC]	OS
labuser-virtual-machine	6/3/2026, 6:40:38.869 A...	Linux
Azure-VM-WIN-SR	6/3/2026, 6:39:49.000 A...	Windows
DESKTOP-NT05UND	6/3/2026, 5:28:34.462 A...	Windows

9.2 Syslog Validation

To verify Linux log ingestion:

KQL Query:

```
Syslog
| take 10
```

Results:

- Logs from Ubuntu VM displayed successfully.
- Included SSH authentication attempts, system startups, and cron jobs.
- Timestamps aligned with local system time (UTC).

Screenshot: Syslog entries from Ubuntu VM visible in Logs.

Microsoft Azure Sentinel-LAW | Logs

Search resources, services, and docs (G+I)

Run Time range: Last 24 hours Show: 1000 results

1 Syslog | take 10

TimeGenerated [UTC]	Computer	EventTime [UTC]	Facility	HostName	SeverityLevel	SyslogMessage	ProcessID	HostIP	ProcessName
> 6/3/2026, 6:05:47:642 AM	labuser-virtual-machine	6/3/2026, 6:05:47:000 AM	daemon	labuser-virtual-machine	warning	Configuration file /run/systemd/system/netplan-ovs-cleanu...	1	127.0.1.1	systemd
> 6/3/2026, 6:05:47:496 AM	labuser-virtual-machine	6/3/2026, 6:05:47:000 AM	authpriv	labuser-virtual-machine	info	new user: name=azuremetricsext, UID=996, GID=996, home...	10218	127.0.1.1	useradd
> 6/3/2026, 6:05:47:299 AM	labuser-virtual-machine	6/3/2026, 6:05:47:000 AM	daemon	labuser-virtual-machine	warning	/lib/systemd/system/snappd.service:23: Unknown key name ...	1	127.0.1.1	systemd
> 6/3/2026, 6:05:47:290 AM	labuser-virtual-machine	6/3/2026, 6:05:47:000 AM	daemon	labuser-virtual-machine	warning	Configuration file /run/systemd/system/netplan-ovs-cleanu...	1	127.0.1.1	systemd
> 6/3/2026, 6:05:47:126 AM	labuser-virtual-machine	6/3/2026, 6:05:47:000 AM	daemon	labuser-virtual-machine	warning	/lib/systemd/system/snappd.service:23: Unknown key name ...	1	127.0.1.1	systemd
> 6/3/2026, 6:05:47:118 AM	labuser-virtual-machine	6/3/2026, 6:05:47:000 AM	daemon	labuser-virtual-machine	warning	Configuration file /run/systemd/system/netplan-ovs-cleanu...	1	127.0.1.1	systemd
> 6/3/2026, 6:05:47:018 AM	labuser-virtual-machine	6/3/2026, 6:05:46:000 AM	authpriv	labuser-virtual-machine	info	new user: name=azuremetriccollector, UID=997, GID=998, ho...	10134	127.0.1.1	useradd
> 6/3/2026, 6:05:46:981 AM	labuser-virtual-machine	6/3/2026, 6:05:46:000 AM	authpriv	labuser-virtual-machine	info	new group: name=azuremonitoragent, GID=998	10128	127.0.1.1	groupadd
> 6/3/2026, 6:05:46:981 AM	labuser-virtual-machine	6/3/2026, 6:05:46:000 AM	authpriv	labuser-virtual-machine	info	group added to /etc/gshadow: name=azuremonitoragent	10128	127.0.1.1	groupadd
> 6/3/2026, 6:05:46:978 AM	labuser-virtual-machine	6/3/2026, 6:05:46:000 AM	authpriv	labuser-virtual-machine	info	group added to /etc/group: name=azuremonitoragent, GID...	10128	127.0.1.1	groupadd

0s 341ms Display time (UTC+00:00) Query details 1 - 10 of 10

Screenshot: SecurityEvents for windows and syslog for ubuntu are working - confirmed
 union SecurityEvent, Syslog
 | where TimeGenerated > ago(30m)
 | summarize count() by Computer

Microsoft Defender Advanced hunting

Selected workspace: Sentinel-LAW Help resources Query resources report

Explore your content from Microsoft Sentinel
 All data from Microsoft Sentinel, including tables, queries and functions is now available for you to explore.

New query* x New query* x

Schema v

Exposure Management

- ExposureGraphEdges
- ExposureGraphNodes

LogManagement

- AADManagedIdentitySigni...
- AADNoninteractiveUserSL...
- AuditLogs
- Event
- Heartbeat
- MicrosoftGraphActivityLogs
- SigninLogs
- Syslog
- Usage

Microsoft Sentinel

- BehaviorAnalytics
- SecurityIncident
- Watchlist
- Cases
- SecurityAlert
- SecurityEvent

Query

Run query

1 union SecurityEvent, Syslog
 2 | where TimeGenerated > ago(30m)
 3 | summarize count() by Computer

Results Query history

Export Show empty columns 2 items Search 0001421 Low

Filters Add filter

Computer count

labuser-virtual-ma... 1745

DESKTOP-NT05UN0 8994

10. Threat Detection Configuration

10.1 Windows Security Auditing

To capture login events, auditing policies were enabled on the **Windows 11 VM**:

- **Audit Logon Events:** Success and Failure
- **Audit Account Logon Events:** Success and Failure
- **Special Logon Auditing:** Enabled

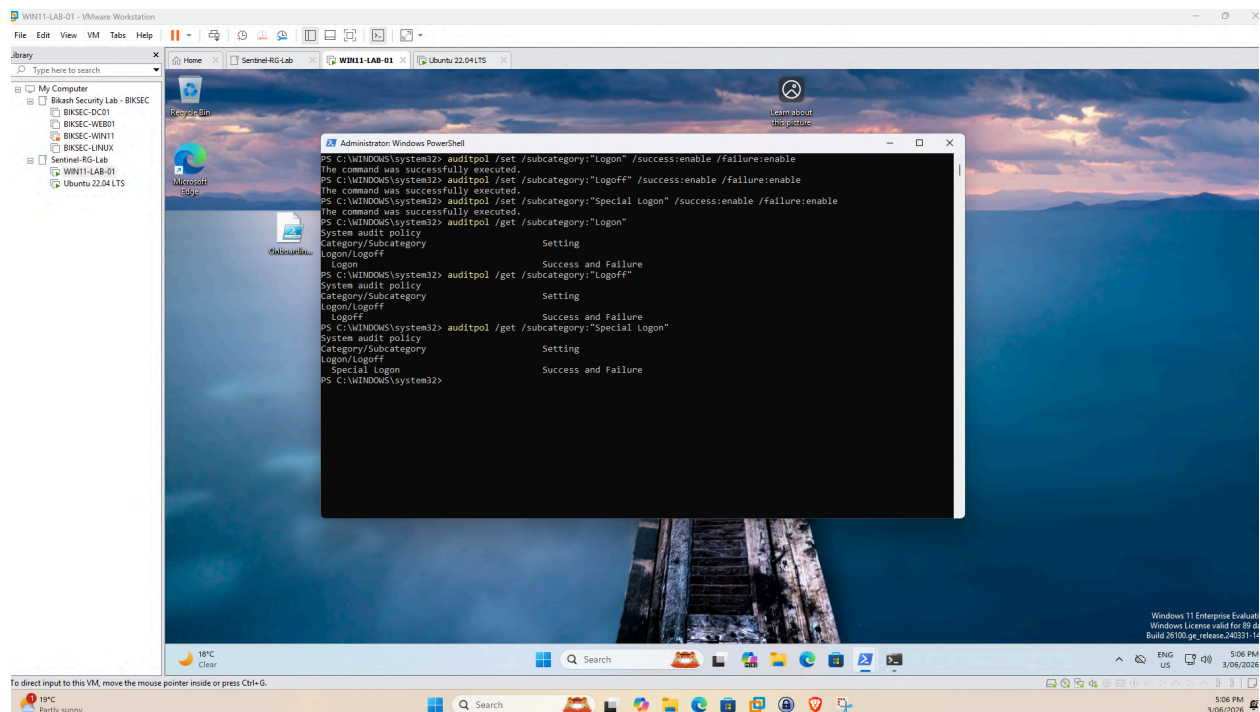
This ensures that **Event ID 4625 (failed logon)** is generated upon failed RDP attempts.

Screenshot: Local Security Policy showing audit settings enabled.

```
auditpol /set /subcategory:"Logon" /success:enable /failure:enable
```

```
auditpol /set /subcategory:"Logoff" /success:enable /failure:enable
```

```
auditpol /set /subcategory:"Special Logon" /success:enable /failure:enable
```



10.2 Failed RDP Login Analytics Rule

A **Scheduled Analytics Rule** was created in Microsoft Sentinel to detect potential brute-force attacks.

Microsoft Defender Portal → Microsoft Sentinel → Analytics → Create → Scheduled rule

Property	Value
Rule Name	Failed_RDP_Logins
Severity	Medium
Tactics	Initial Access
Technique	T1110 – Brute Force
Query	See below
Query Frequency	Every 5 minutes
Lookback Period	5 minutes
Threshold	>3 failed attempts per user/computer
Incident Creation	Enabled

Detection Query:

```
Event
| where EventID == 4625
| extend Account = tostring(parse_json(EventData).TargetUserName)
| summarize FailedAttempts = count() by Account, Computer
| where FailedAttempts > 3
```

Screenshot: Analytics rule creation form with query and settings.

Microsoft Defender | Default Directory

Search > Analytics

Manage all your rules in one place

Now you can manage all your rules on one page, providing a centralized and streamlined approach to rule management. This enhancement not only simplifies your workflow but also ensures that you can easily access, modify, and oversee all your rules in one convenient location.

Go to unified rules page

Active rules Rule templates Anomalies

Create Analytics workbooks Rule runs (Preview) Enable Disable Delete Import Export Columns

Scheduled query rule

No analytics rules were found

What is it?

Analytics rules provide you the ability to detect and respond to threats in your organization. [Learn more](#)

How does it work?

You can create a [new analytics rule](#) or enable a rule from our gallery of built-in [rule templates](#)

Make the most out of analytics rules

Reduce noise

Carefully define the threats you want to investigate in your environment. Control the threshold, severity, and more.

Map entities

When you define analytics rules, make sure to map important entities. This enables advanced investigation options when that alert is triggered.

Enable built-in analytics rules

Microsoft Sentinel provides a rich set of built-in rule templates that you can enable and customize.

Respond to alerts automatically

Define automated Playbooks and associate them with your analytics rules, so they run automatically when an alert is triggered.

18°C Mostly sunny 12:10 PM 4/06/2026

Microsoft Defender | Default Directory

Search > Analytics > Analytics rule wizard

Analytics rule wizard - Create a new Scheduled rule

General

Set rule logic

Incident settings

Automated response

Review + create

Create an analytics rule that will run on your data to detect threats.

Analytics rule details

Name *

Failed_RDP_Logins

Description

Severity

Medium

MITRE ATT&CK

2 Selected

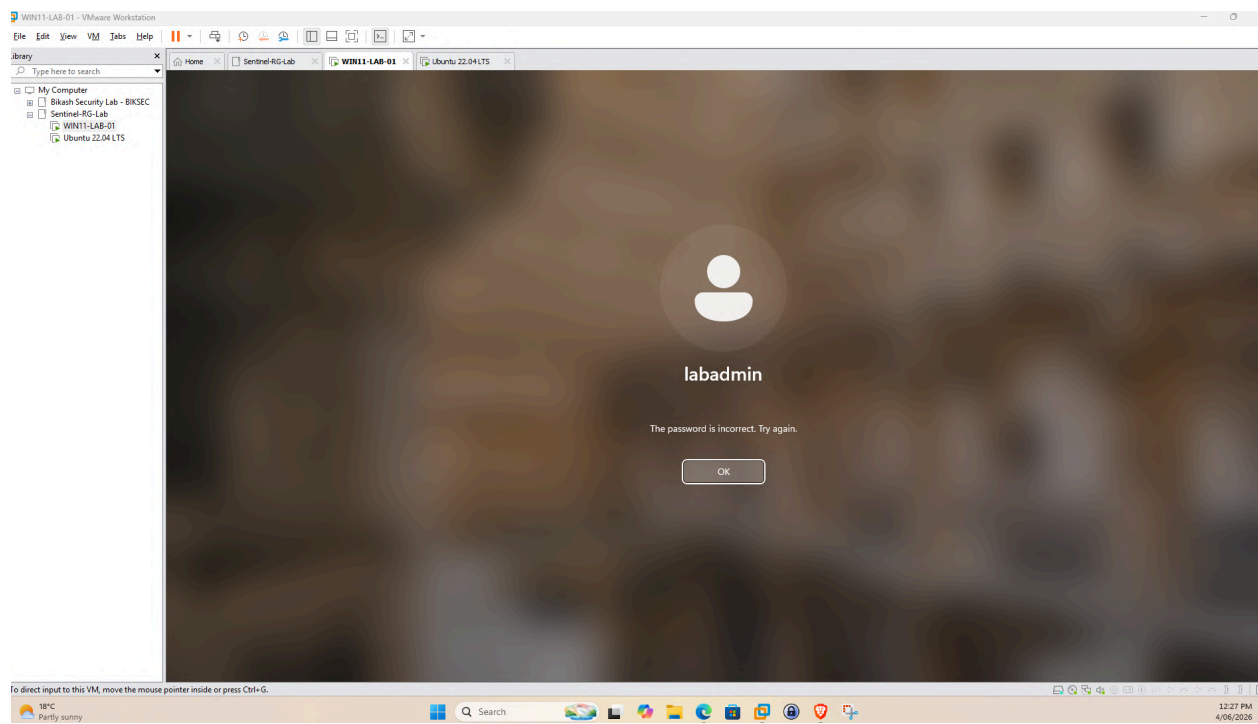
Status

Enabled

Next: Set rule logic >

Cancel

18°C Mostly sunny 12:12 PM 4/06/2026



Screenshot: Alert generated in Microsoft Sentinel.

Alerts - Microsoft Defender

https://security.microsoft.com/alerts?tid=ea466e5-0a7b-4473-b5c0-e9a9c6f6a255

Microsoft Defender

Find Defender solutions

Home

Exposure management

Investigation & response

Incidents & alerts

Alerts

Hunting

Actions & submissions

Partner catalog

Threat intelligence

Assets

Microsoft Sentinel

Search

Threat management

Content management

Configuration

Tables

Data connectors

Analytics

Summary rules

Watchlist

Automation

Email & collaboration

Cloud security

Cases

SOC optimization

Reports

Alerts

Alert service settings

Loading...

Built-in alert tuning was updated with new rules. These rules help streamline your incident and alert queues by prioritizing high-fidelity, actionable alerts. Review or manage the affected alert types in [Alert Tuning](#). [Learn more about built-in rules.](#) [Don't show again](#)

Export 1 Week 3 Alerts Search for name or ID Customize columns

Filter set: Status: New, In progress Add filter Reset all

Alert name	Tags	Severity	Investigation state	Status	Category	Detection source	Product name	Impacted assets
Failed_RDP_Logins		Medium		New	Credential Access	Scheduled detection	Microsoft Sentinel	
Failed_RDP_Logins		Medium		New	Credential Access	Scheduled detection	Microsoft Sentinel	
Failed_RDP_Logins		Medium		New	Credential Access	Scheduled detection	Microsoft Sentinel	

Microsoft Defender

Find Defender solutions

Home

Exposure management

Investigation & response

Incidents & alerts

Alerts

Hunting

Actions & submissions

Partner catalog

Threat intelligence

Assets

Microsoft Sentinel

Search

Threat management

Content management

Configuration

Tables

Data connectors

Analytics

Summary rules

Watchlist

Automation

Email & collaboration

Cloud security

Cases

SOC optimization

Reports

Alerts > Failed_RDP_Logins

Loading...

Part of incidents: [Failed_RDP_Logins](#) [View incident page](#)

What happened

ANALYTICS RULE

Analytics rule details

Rule name: Failed_RDP_Logins [View rule in Sentinel](#)

Rule description

Related events

Query results

View query 1 item Customize columns

Account	Computer	FailedAttempts
Computer	DESKTOP-NTOSLUND	5

Failed_RDP_Logins

Medium Unknown New

Manage alert Move alert to another incident

Set Classification

Alert details

Alert ID: sn81a04a68-464a-4396-9c76-51b172e565df Categories: Credential Access

Detection source: Scheduled detection Service source: Microsoft Sentinel

Detection technology: Generated on: Jun 5, 2026 10:46:07 AM

First activity: Jun 5, 2026 10:35:41 AM Last activity: Jun 5, 2026 10:40:41 AM

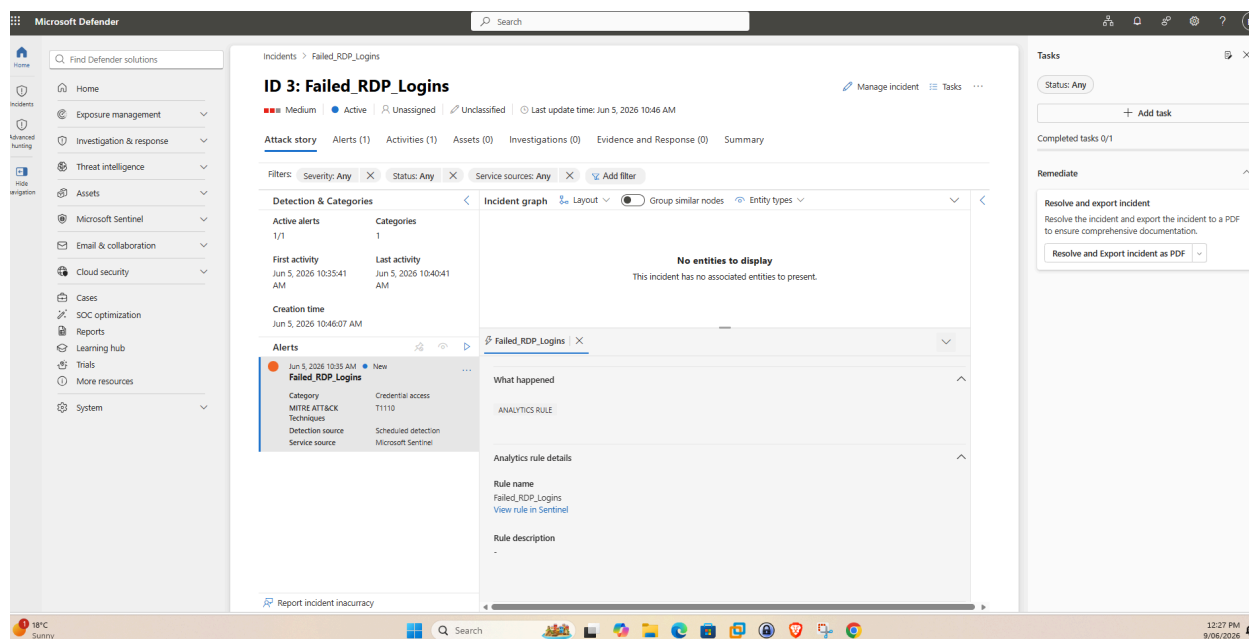
Workspace: Sentinel-LAW

Evidence

No related evidence was found

Analytics rule details

Rule name Rule description



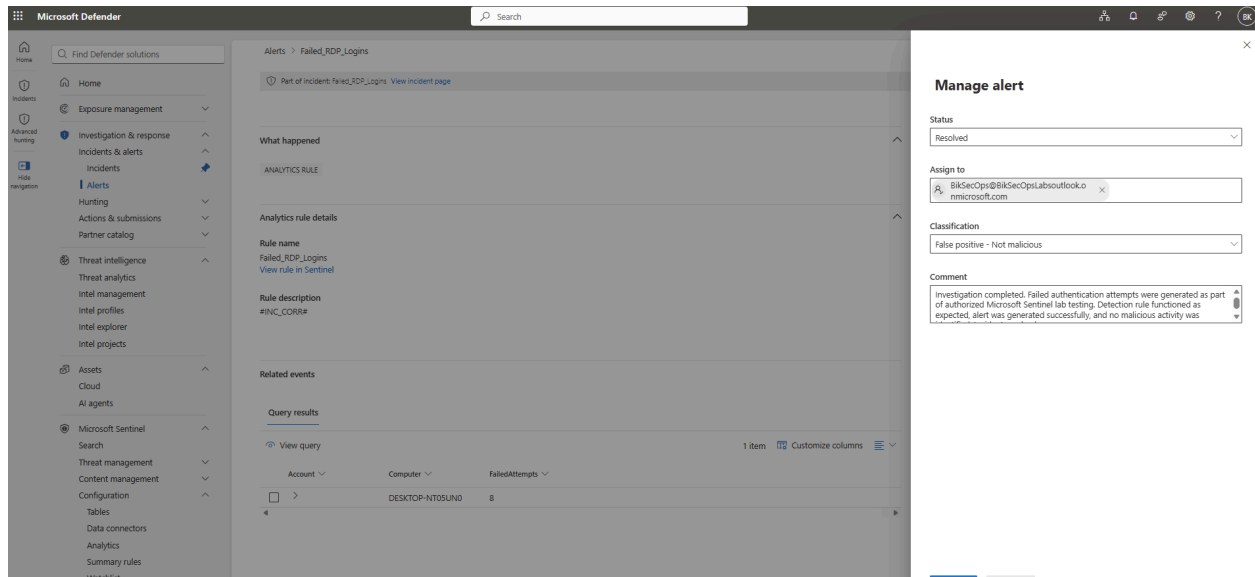
12. Incident Response

The generated incident was assigned for investigation within Microsoft Sentinel. After reviewing the associated Event ID 4625 authentication failures, it was confirmed that the activity was generated as part of authorized lab testing.

An investigation comment was added, and the incident status was changed to **Resolved**.

Investigation Comment:

Investigation completed. Failed authentication attempts were generated as part of authorized Microsoft Sentinel lab testing. Detection rule functioned as expected, alert was generated successfully, and no malicious activity was identified. Incident resolved.



In a production environment, the incident would be escalated and handled according to the organization's incident response playbooks. However, as the activity was confirmed to be legitimate testing, the incident was documented and marked as **Resolved**.

13. Results

All project objectives were met with full success:

Objective	Status	Notes
Deploy Microsoft Sentinel	✓	Enabled on Log Analytics Workspace
Create Log Analytics Workspace	✓	Named Sentinel-LAW
Configure DCE and DCR	✓	Secure data ingestion pipeline established
Implement RBAC	✓	Least-privilege roles assigned
Onboard on-prem Windows/Linux	✓	Via Azure Arc
Onboard Azure VM	✓	Direct integration
Install AMA	✓	All endpoints reporting
Validate telemetry	✓	Heartbeat and Syslog confirmed
Centralized logging	✓	Unified visibility across hybrid estate

Create analytics rule	✓	Detects RDP brute-force attempts
Generate alerts/incidents	✓	End-to-end detection validated
Investigate and Resolve Incident	✓	Incident assigned, reviewed, and resolved

14. Conclusion

This project successfully implemented a hybrid Microsoft Sentinel SIEM/SOC laboratory environment. Windows and Linux endpoints were onboarded using Azure Arc, while an Azure-hosted Windows Server was integrated directly into the monitoring environment. Telemetry collection was validated through Heartbeat and Syslog logs, confirming successful connectivity and data ingestion.

A custom Failed RDP Login detection rule was implemented to demonstrate security monitoring and threat detection capabilities. The successful generation of alerts and incidents confirmed that Microsoft Sentinel was operating effectively as a centralized SIEM platform capable of monitoring, detecting, and responding to security events across a hybrid infrastructure.